

CSCE 411: Mastery Quiz 2

Name: _____

UIN: _____

Let f_1, f_2, f be functions from $\mathbb{Z}^+$ to $\mathbb{R}_{\geq 0}$ such that $f_1(n) \in \Omega(f(n))$ and $f_2(n) \in \Theta(f(n))$. Prove that $f_1(n) \cdot f_2(n) \in \Omega(f(n)^2)$.

Solution: Let $c_1, c_2 \in \mathbb{R}^+$, $n_1, n_2 \in \mathbb{Z}^+$ be constants such that $\forall n \geq n_1, f_1(n) \geq c_1 f(n)$ and $\forall n \geq n_2, f_2(n) \geq c_2 f(n)$ (such constants must exist by the definitions of big- Ω and big- Θ). To show that $f_1(n) \cdot f_2(n) \in \Omega(f(n)^2)$ we choose the constants $c = c_1 \cdot c_2$ and $n_0 = \max(n_1, n_2)$. Then, by our choice of c_1, c_2, n_1 , and n_2 , we have that for every $n \geq n_0$

$$f_1(n) \cdot f_2(n) \geq c_1 f(n) \cdot c_2 f(n) = c f(n)^2.$$

So by the definition of big- Ω , $f_1(n) \cdot f_2(n) \in \Omega(f(n)^2)$.